

Sr.No	Module	Weightage %	Marks Weightage%
1	Module-III	20.00	31.58
2	Module-IV	20.00	26.32
3	Module-II	20.00	19.74
4	Module-I	20.00	14.47
5	Module-V	20.00	7.89

Weightage representation okay : Weightage representation exceeded more than 5% :Weightage representation exceeded more than 10%

CSE601

[ET]

Enroll. No. - _____

END SEMESTER EXAMINATION : April-May, 2026

Data Structures and Algorithm Design

Time : 3.00 Hrs

Maximum Marks : 60

Note: Attempt questions from all sections as directed.

Section - A : Attempt any Four questions out of Five . Each question carries 06 marks.

- Q1.

(a)

Define Big-oh, Big-Omega and Big-Theta.

(2)

Delete

(b)

Define push() and pop() operations.

(2)

Delete

(c)

Write the time complexity of Stack and Queue Operations.

(2)

Delete
- Q2.

(a)

Define the Binary Search Algorithm.

(2)

Delete

(b)

Convert Postfix expression of the expression -1+B-C+D and A+B*(C-D)/(P-R)

(2)

Delete

(c)

Define enqueue() and dequeue() operations.

(2)

Delete
- Q3.

(a)

Define Splay Tree.

(2)

Delete

(b)

Define Red-Black Tree.

(2)

Delete

(c)

What are the advantages of Dynamic Programming?

(2)

Delete
- Q4.

(a)

What is the main idea of Quick Sort?

(3)

Delete

(b)

Define String Matching Algorithm and Approximation Algorithm.

(3)

Delete
- Q5.

What are NP, NP-Complete and NP-Hard Problems? Give examples.

(6)

Delete

Section – B : Attempt any two questions out of three. Each question carries 10 marks.

- Q6.

(a)

Write a complete Java Program for Doubly Linked List InsertAtBegin, DeleteFromBegin, and Display operations.

(5)

Delete

(b)

Discuss the Quick sort algorithm analysis with example data.

(5)

Delete
- Q7.

(a)

Write Java Code for Level Order Traversal in a Binary Tree.

(5)

Delete

(b)

Write Java Code to insert an operation in a BST (Binary Search Tree).

(5)

Delete
- Q8.

(a)

Write a complete Java Program for Doubly Linked List InsertAtBegin, DeleteFromBegin, and Display operations.

(5)

Delete

(b)

Discuss the Quick sort algorithm analysis with example data.

(5)

Delete

Section - C : Compulsory question

- Q9.

(a)

Write the Time Complexity of all the sorting algorithms.

(4)

Delete

(b)

Discuss different ways to represent Graph Data Structure.

(4)

Delete

(c)

Write the BFS and DFS algorithms. Give their time complexity.

(4)

Delete

(d)

Discuss Recursion vs Dynamic Programming.

(4)

Delete